B.TECH. PROJECT -2

ON

THE INTEGRATION OF GPS AND TELEMETERY IN QUADCOPTER DRONE

Report submitted in partial fulfillment of the requirements for the B. Tech. degree in Mechanical Engineering (Electric Vehicles)

Submitted By

NAMAN DAYAL	**2021UMV7603**
HRITESH GAUTAM	**2021UMV7656**
SHREYA NARAYANAN	**2021UMV7659**
PRERIT SRIJAN	**2021UMV7676**

Under the Guidance
of
Mr. Manish Kumar
Assistant Professor

DEPARTMENT OF MECHANICAL ENGINEERING (ELECTRIC VEHICLES)

NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY (NSUT), WEST CAMPUS
JAFFARPUR KALAN - 110073, NEW DELHI

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CANDIDATE'S DECLARATION

DEPARTMENT OF MECHANICAL ENGINEERING (ELECTRIC VEHICLES)

I/We, Naman Dayal (2021UMV7603), Hritesh Gautam (2021UMV7656), Shreya Narayanan (2021UMV7659), Prerit Srijan (2021UMV7676), students of B. Tech., Department of Mechanical Engineering, hereby declare that the Project-Thesis titled "THE INTEGRATION OF GPS AND TELEMETERY IN QUADCOPTER DRONE " which is submitted by us to the Department of Mechanical Engineering (Electric Vehicles), Netaji Subhas University of Technology, West Campus, Jaffarpur, New Delhi in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology is my/our original work and not copied from any source without proper citation. The manuscript has been subjected to plagiarism checks by Turnitin software. This work has not previously formed the basis for the award of any other Degree.

Place:

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NAMAN DAYAL	HRITESH GAUTAM	SHREYA NARAYANAN	PRERIT SRIJAN
2021UMV7603	2021UMV7656	2021UMV7659	2021UMV7676

CERTIFICATE OF DECLARATION

DEPARTMENT OF MECHANICAL ENGINEERING (ELECTRIC VEHICLES)

This is to certify that project title "THE INTEGRATION OF GPS AND TELEMETERY IN QUADCOPTER DRONE" which is being submitted by Naman Dayal (2021UMV7603), Hritesh Gautam (2021UMV7656), Shreya Narayanan (2021UMV7659), Prerit Srijan (2021UMV7676) to Department of Mechanical Engineering(EV), Netaji Subhas University of Technology, West Campus, Jaffarpur, New Delhi-110073 in partial fulfillment of the requirement of the degree of Bachelors of Technology, is a record of report work carried out by the students under my supervision and guidance. To the best of my knowledge, the content of this report, in full or in any parts, has not been submitted for any other degree or diploma.

Place:

Date:

Mr. Manish Kumar

Assistant professor

Department of Mechanical Engineering

Netaji Subhas University of Technology, West Campus

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Place:

Date:

NAMAN DAYAL	HRITESH GAUTAM	SHREYA NARAYANAN	PRERIT SRIJAN
2021UMV7603	2021UMV7656	2021UMV7659	2021UMV7676

ABSTRACT

The integration of GPS and telemetry systems in quadcopter drones represents a significant advancement in the realm of unmanned aerial vehicle (UAV) technology. This paper explores the synergistic combination of these two technologies, focusing on their role in improving the navigation, control, and monitoring of quadcopters. The GPS system provides precise location tracking and navigation capabilities, while telemetry enables real-time data transmission, allowing for continuous monitoring of the drone's status, including altitude, speed, battery life and system health. Through a detailed examination of the integration process, the paper highlights the benefits, such as enhanced flight stability, improved mission accuracy and more efficient route planning. Furthermore, the study discusses the challenges encountered, as well as the potential solutions to overcome these obstacles. By synthesizing the strengths of GPS and telemetry, this research contributes to the development of more efficient, reliable, and intelligent quadcopter drones for a wide range of applications, including aerial surveying, search and rescue, and environmental monitoring.

Keywords: GPS, Telemetry, Quadcopter Drones, Navigation, Control, Monitoring, Autonomous Operations, Flight Stability, Mission Accuracy, System Reliability

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LIST OF NOMENCLATURE AND ABBREVIATIONS

UAV: Unmanned Aerial Vehicle

VTOL: Vertical Take-Off and Landing

BLDC: Brushless Direct Current

ESC: Electronic Speed Controller

GPS: Global Positioning System

UAV: Unmanned Aerial Vehicle

LIPO: Lithium Polymer

PWM: Pulse Width Modulation

Rx: Radio System Receiver

C.G.: Center of Gravity

KV: Motor Speed Constant (RPM/Volt)

BEC: Battery Elimination Circuit

T: Torque (Nm)

I: Current (A)

Kt: Torque Constant (Nm/A)

mAh: Milliampere-hour

N: Motor Speed (RPM)

D: Propeller Diameter (m)

P: Propeller Pitch (m)

ρ: Air Density (kg/m^3)

v: Air Velocity (m/s)

Fc: Centrifugal Force (N)

ω: Angular Speed (rad/s)

M: Total Mass Lifted by Quadcopter (kg)

W: Weight (N)

Y: Propeller Length (cm)

Chapter 1: Introduction

Background

A quadcopter is an aircraft that exceeds air in weight and can perform vertical take-off and landing (VTOL). It is driven by four rotors situated on the same level, parallel to the ground. Unlike conventional helicopters, quadcopters employ fixed-pitch blades in their rotors and navigate through the air by adjusting the relative speed of each propeller. The pioneering quadcopter, the Omnichen2, was created in 1920 by Etienne Omni Chen. This aircraft completed 1000 successful flights and covered a documented distance of 360 meters. Subsequently, the convert a wing model, a quadcopter designed by Dr. George E. Bothesate, emerged in 1956. The 21st century has witnessed remarkable advancements in quadcopter technology. Universities, students, and researchers are continuously developing more sophisticated controllers and modelling techniques to provide precise and accurate representations of real-world quadrotors. Consumers can now purchase quadcopters of various sizes from department stores and specialized retailers, though prices for these aerial vehicles remain high. This article aims to provide information on the design, construction, and demonstration of a quadcopter capable of performing diverse tasks across multiple fields, enabling students to build quadcopters at home at a significantly lower cost. We can categorize quadcopters into two groups: micro and mini air vehicles, based on their size and weight. We have utilized two pairs of identical fixed-pitch propellers, with two rotating clockwise (CW) and two counterclockwise (CCW). Control is achieved by independently varying the speed of each rotor. By altering individual rotor speeds, it is possible to generate a specific total thrust, position the centre of thrust both laterally and longitudinally, and create a desired total torque or turning force. Quadcopters differ from traditional helicopters, which use rotors capable of dynamically adjusting their blade pitch as they revolve around the rotor hub.

Modern quadcopter drones are increasingly being designed to operate autonomously and transmit real-time data to a remote user. Two critical technologies that make this possible are GPS (Global Positioning System) and telemetry systems. The integration of these technologies provides the drone with enhanced navigation, control, and communication capabilities, making it suitable for complex tasks across industries such as agriculture, surveillance, environmental monitoring, and delivery services.

1. Role of GPS in a Quadcopter Drone

The GPS system in a drone is responsible for determining the drone's precise geographical position, altitude, velocity, and time by receiving signals from satellites orbiting the Earth. Here's what GPS enables in a drone:

- Waypoint navigation: Drones can follow pre-defined routes with GPS coordinates.
- Return to Home (RTH): If the drone loses connection or runs low on battery, it can return to the launch point automatically.
- Geofencing: Users can create virtual boundaries that the drone cannot cross.
- Precision landing: GPS data helps drones land at the exact take off location.

2. Role of Telemetry in a Quadcopter Drone

Telemetry is the wireless communication link between the drone and the ground control station (GCS). It typically uses radio frequency (RF) modules, Wi-Fi, or cellular networks to transmit data such as:

- Real-time flight data (altitude, speed, orientation)
- GPS coordinates
- Battery voltage and system health
- Sensor readings (IMU, barometer, compass)
- Live mission updates and feedback

Telemetry also allows users to send commands from the ground to the drone, such as changing the flight path or activating a payload (e.g., a camera or sprayer).

3. Integration and Functionality

When GPS and telemetry are integrated:

- The GPS module collects positioning data.

- This data is sent via the flight controller to the telemetry module.
- The telemetry system transmits this data to the ground station in real time.
- The ground station software (e.g., Mission Planner, Ground Control) visualizes the drone's location on a map and allows operators to control or adjust missions dynamically.

This integration enables:

- Autonomous missions: Drones can fly without manual control, following GPS-defined routes.
- Real-time monitoring: Operators can observe drone status and location continuously.
- Data logging: Flight data is recorded for analysis and reporting.
- Safety and recovery: In emergencies, GPS + telemetry helps recover lost drones or prevent crashes.

1.1 Project Motivation

The swift advancement of unmanned aerial vehicle (UAV) technology has transformed various industries, including agriculture, supply chain management, emergency response, ecological surveillance, and military operations. UAVs provide diverse applications, ranging from precise agricultural practices and parcel transportation to surveillance and search-and-rescue missions. Nevertheless, enhancing the design, productivity and functionality of UAVs remains a crucial focus in ongoing research and innovation efforts. This initiative, which concentrates on UAV design and construction, as well as propeller thrust evaluation, is driven by the necessity to tackle these issues and further the progress of unmanned aerial systems (UAS).

Key Motivations

1. Understanding the Fundamentals of UAV Design

Unmanned aerial vehicles, commonly known as drones, represent a sophisticated

fusion of mechanical, electrical, and software engineering. This endeavor offers a chance to explore in- depth the fundamental concepts of fluid dynamics, structural engineering, and propulsive systems. Through the process of conceptualizing and constructing a drone, we seek to broaden our comprehension of the interplay between various elements—including the chassis, engines, rotors, and energy systems—that collectively contribute to achieving stable and optimized flight performance.

2. Thrust Optimization for Enhanced Performance

The thrust generated by propellers plays a crucial role in determining a drone's ability to lift, maintain stability, and carry payloads. Our objective is to conduct comprehensive thrust tests to determine the ideal design specifications, including the angle of blade pitch, propeller length and material composition. These tests aim to maximize efficiency while reducing power usage, which is especially critical for applications that require extended flight times and increased cargo-carrying capabilities.

3. Bridging Theory and Practical Application

While theoretical knowledge provides the foundation for innovation, practical implementation is essential for understanding real-world challenges. This project offers hands-on experience in:

1. **Material selection** for lightweight and durable drone components. **Precision manufacturing** using tools like CNC machining or 3D printing for propeller and frame fabrication.
2. **Performance testing** to validate design assumptions.

4. Addressing Industry Needs

The drone industry is experiencing rapid growth, with a rising demand for customized solutions tailored to specific applications. By designing a drone from the ground up and performing an in-depth thrust analysis, we aim to provide innovative insights that can be applied to practical, real-world scenarios.

5. Sustainability and Efficiency

This project is further driven by the growing need for energy-efficient drones, particularly in the context of renewable energy and sustainable aviation. By

focusing on thrust testing and optimization, we aim to significantly lower energy consumption, making drones both more eco-friendly and cost-efficient.

6. Skill Development for Future Technologies

Engaging in this project will help us develop a multidisciplinary skill set encompassing CAD modeling, structural analysis, aerodynamic testing, and control system integration. These competencies are essential for careers in aerospace, robotics, and mechatronics, positioning us as valuable contributors to future technological innovations.

1.2 Broader Impact

The knowledge gained from this project could have significant implications:

- Improving the reliability of drones used in **disaster response**, where stability and payload capacity are critical.
- Enhancing drone efficiency for **last-mile delivery**, reducing operational costs for industries.

Providing insights into the design of drones for **precision agriculture**, enabling better crop monitoring and resource management.

1.3 Structure of the Report

The report begins with a Title Page, which includes the title "Design and Fabrication of Quadcopter Drone," along with the names and roll numbers of the contributors, the supervisor's details, and the name of the department and university. It also states the purpose of the report as being submitted in partial fulfillment of the requirements for the B.Tech degree. The title page also includes the date of submission.

The next section is the Candidate’s Declaration, where the contributors declare that the work submitted is original, has undergone plagiarism checks, and has not been submitted elsewhere for any other degree or diploma. This section includes the signatures of all contributors to affirm the declaration.

Following this is the Certificate of Declaration, a certification from the supervisor verifying that the report has been completed under their guidance. It confirms the authenticity of the work and provides the supervisor's signature and designation.

The Acknowledgment section expresses gratitude to the contributors, including faculty

members, colleagues, and laboratory staff, for their guidance, support, and encouragement in completing the project. It recognizes the assistance received during the course of the work.

The Abstract provides a concise summary of the project, outlining its objectives, methodology, and results. It highlights the significance of the work and provides a quick overview of the report for the reader.

A list of Keywords follows, including important terms related to the project, such as UAV, quadcopter, BLDC motors, GPS , Telemetery and other key concepts that define the scope of the report.

The Index contains a detailed table of contents with page numbers, followed by the lists of figures and tables. This section helps readers navigate through the report efficiently.

The Nomenclature and Abbreviations section defines the technical terms, symbols, and abbreviations used throughout the report. This ensures that readers understand the terminology, enhancing the clarity of the document.

Chapter 2: Literature Review and Problem Statement

2.1 Literature Review

Drones emerged in the twentieth century as technological innovations designed to perform specific tasks, aiming to minimize human involvement. Their design is a critical factor influencing their development and overall functionality. Nithyavathy et al. (2020) and Tatale et al. (2019) have made significant contributions to drone design, with extensive research conducted in this area. Quadcopter development has evolved from early experimental designs to modern UAVs with enhanced stability, manoeuvrability, and payload capacity, thanks to advancements in brushless motors, electronic speed controllers, and microcontrollers. Recent innovations have improved control mechanisms and material selection, enabling applications in surveillance, transportation, and industrial automation as presented by Tatale et al. (2018). Topology optimization has been extensively applied to drone frame design, enhancing structural integrity, weight reduction, and aerodynamic efficiency. Recent studies demonstrate a 30% reduction in weight while maintaining performance, improving agility, energy efficiency, and maneuverability for future UAV applications as presented by (Bay & Eryıldız, 2024). The research by Yemle et al. (2019) presents a versatile drone frame designed for both quadcopters and octocopters. By utilizing aluminium pipes and 3D-printed joints, the study aims to reduce weight and manufacturing costs. Structural analysis in ANSYS confirms durability, supporting higher payloads for various industrial applications. Šustek et al. (2018) explored quadcopter components, including propellers, motors, ESCs, batteries, and flight controllers. They highlight the influence of propeller design, BLDC motor efficiency, and ESC operations. The study also evaluates LiPo battery performance and compares Arduino and Raspberry Pi controllers for UAV automation and flight control. Priandana et al. (2020) developed autonomous UAV quadcopters using Pixhawk controllers for smart farming. The research covers component assembly, sensor calibration, flight path planning, and data collection. Results confirm accurate autonomous flights with minimal deviations, demonstrating UAV potential in agriculture for improved automation, efficiency, and precision in various applications. Early developments in quadcopter technology date back to the early 20th century, with significant advancements in flight stability achieved through improved sensors and microcontrollers. Recent innovations have enhanced their

capabilities, integrating GPS, infrared sensors, and advanced control systems for applications in surveillance, agriculture, and autonomous navigation as presented by Ghosh et al. (2018). Supriyono et al. (2021) explored the effectiveness of combining GPS and computer vision to improve the landing precision of quadcopter drones. Their research involved a comparative analysis of two autonomous landing approaches: one using only GPS and the other integrating a computer vision algorithm alongside GPS. Flight tests were conducted at an altitude of 5 meters to evaluate the accuracy of each method. The study provides valuable insights into enhancing drone navigation and precision landing capabilities. The paper discusses the development of a programmable autonomous quadcopter (PAQ) designed for swarm robotics research. The authors modified a CrazyFlie 2.0 by adding sensors and revising the software to enable autonomous flight, addressing challenges like sensor noise, weight imbalance, and battery voltage drops. The paper also details the testing of the PAQ's speed and battery life as presented by Kamrath et al. (2017). Extensive work has been done on the design and experimental validation of the Peregrine Falcon, a foldable and launchable quadcopter drone capable of autonomous deployment and stabilization after barrel launching as presented by Wu et al. (2024). Another research explored short-range telemetry communication for autonomous drone navigation, leveraging HC-05 Bluetooth modules to interface a Raspberry Pi companion computer with a Pixhawk Mini autopilot for indoor testing and image processing applications as presented by Anand M et al. (2020).

2.2 Key challenges and approaches to resolve problems

Integrating GPS and telemetry into a quadcopter drone involves various technical and practical challenges. These challenges can affect flight stability, communication reliability, data accuracy, and overall drone performance. Identifying and addressing these issues is crucial for building an effective and robust system. The following sections outline the major challenges and recommended approaches to resolve them.

1. GPS Signal Accuracy and Reliability

Challenge:

- GPS signals can be weak or lost in areas with obstructions (e.g., buildings, trees, tunnels).

- Multipath errors and atmospheric conditions can reduce location accuracy.
- Delay in satellite signal acquisition at startup.

Approaches to Resolve:

- Use High-Sensitivity GPS Modules: Select GPS modules with multi-constellation support (e.g., GPS + GLONASS + Galileo) to improve reliability and reduce signal loss.
- Install GPS Away from Interference: Mount the GPS module away from power lines and ESCs (Electronic Speed Controllers) to reduce electromagnetic interference.
- Use GPS with Compass (GNSS + Magnetometer): Helps in improving heading accuracy, especially in stationary mode.
- Enable GPS Fusion Algorithms: Use flight controllers (like Pixhawk) that support GPS-IMU data fusion (EKF – Extended Kalman Filter) for better positioning.

2. Telemetry Range and Signal Loss

Challenge:

- Limited communication range due to low-power RF modules.
- Signal loss during long-range flights or in high-interference environments.
- Low data rate affecting real-time control and monitoring.

Approaches to Resolve:

- Use Long-Range Telemetry Modules: Implement high-power RF modules (e.g., 915 MHz, 433 MHz) or switch to 4G/LoRa-based telemetry systems for extended range.
- Proper Antenna Placement: Ensure antennas are mounted vertically and free from obstructions to maximize signal strength.

- Use Error Correction Protocols: Implement protocols like MAVLink that support data packet verification and retransmission.

3. Power Consumption and Battery Drain

Challenge:

- GPS and telemetry modules continuously draw power, reducing flight time.
- Poor power management can cause voltage drops affecting GPS lock or telemetry disconnection.

Approaches to Resolve:

- Efficient Power Distribution: Use a Power Distribution Board (PDB) with voltage regulation (BECs) for stable module power.
- Optimize Power Usage: Use modules with low-power modes and only activate high-frequency telemetry when necessary.
- Battery Monitoring System: Integrate telemetry-based battery voltage monitoring to alert users of low power.

4. Data Synchronization and Latency

Challenge:

- Delay or mismatch between GPS position updates and telemetry transmission can affect navigation accuracy.
- High latency in telemetry leads to delayed commands and unstable control.

Approaches to Resolve:

- Use Real-Time Flight Controllers: Select controllers with fast processors (e.g., STM32-based Pixhawk) that handle GPS/IMU/telemetry data efficiently.
- Implement Time Stamping and Buffering: Ensure data packets are timestamped and buffered correctly for accurate logging and command execution.

- Optimize Baud Rate: Match telemetry module baud rate (typically 57600 or 115200) with the controller to reduce communication delay.

5. Interference Between Modules

Challenge:

- Electromagnetic interference (EMI) between GPS, telemetry, motors, and ESCs can degrade performance.
- Frequency overlap (e.g., GPS at 1.575 GHz and telemetry near 900 MHz) may cause noise.

Approaches to Resolve:

- Shield Sensitive Components: Use EMI shielding tape or cases around GPS modules.
- Separate Wiring Paths: Route signal and power wires separately to reduce noise.
- Use Filters: Add LC filters or ferrite beads to telemetry and GPS power lines.

6. Software Integration Issues

Challenge:

- Improper configuration of flight control software (e.g., ArduPilot or PX4) can lead to failure in GPS locking or telemetry connection.
- Compatibility issues between modules and firmware versions.

Approaches to Resolve:

- Use Standardized Protocols (e.g., MAVLink): Ensure all communication follows standard message structures.
- Regular Firmware Updates: Keep GPS modules, telemetry modules, and flight controllers updated with the latest firmware.
- Simulate and Test in GCS: Use tools like Mission Planner or qgroundcontrol to simulate missions and test GPS/telemetry configurations before flight.

7. Environmental and Safety Factors

Challenge:

- Weather conditions like heavy cloud cover or electromagnetic storms affect GPS reception.
- Loss of telemetry in critical missions could result in drone crashes or flyaways.

Approaches to Resolve:

- Fail-Safe Mechanisms: Program RTH (Return-to-Home) and land-on-loss-of-signal (LoS) features.
- Data Logging: Store telemetry and GPS logs onboard for post-flight analysis.
- Use Redundant Systems: Add backup communication channels or dual GPS modules for redundancy.

Chapter 3: Integration of GPS and Telemetry

3.1 ReadytoSky M8N GPS module

This module is a small device that helps drones and robots figure out their exact location using satellites. It works with multiple satellite systems to improve accuracy and includes a built-in compass for better navigation. It's designed to use little power and quickly find signals, making it great for outdoor projects.



Figure 3.1 ReadytoSky M8N GPS module

3.2 Integration of GPS module on quadcopter drone

Step 1: Open all the propellers.

Step 2: Install the GPS stand in the forward direction of the flight controller on the left side.

Step 3: Install the M8N Ready to sky GPS module on the stand using double sided tape.

Step 4: Tie the wire of the GPS module with the stand with the help of zip ties.

Step 5: Do the configuration of GPS.



Figure 3.2 GPS mounting stand

3.3 3DR Radio Telemetry 915 MHz

This 3DR 915 MHz Radio Telemetry kit set allows you to connect a flight controller and a USB or UART-enabled device, such as a computer, laptop, or tablet with USB compatibility (OTG).

The latest version 2 radios no longer require a separate Air/Ground module, as both transceivers can now use both UART or USB connections, and support both modes.

This radio set not only allows you to view real-time data, such as live GPS position overlaid on a map, system voltage, heading, waypoint navigation, and even an artificial horizon, but also offers much more.

By using an open-source MAVlink-based ground station software, you can even fly your aircraft with a joystick via your computer and update settings in real time. Alternatively, you can use it for your DIY projects that require a wireless Serial interface.

This kit is cost-effective, has a longer range, and offers superior performance. The system operates on the 915 MHz band and provides a full-duplex link using HopeRF's HM-TRP modules running custom, open-source firmware.



Figure 3.3 3DR Radio Telemetry 915 MHz

3.4 Integration of telemetry kit on quadcopter drone

Step 1: Connection of Radio Telemetry 915 MHz to the Pixhawk flight controller.

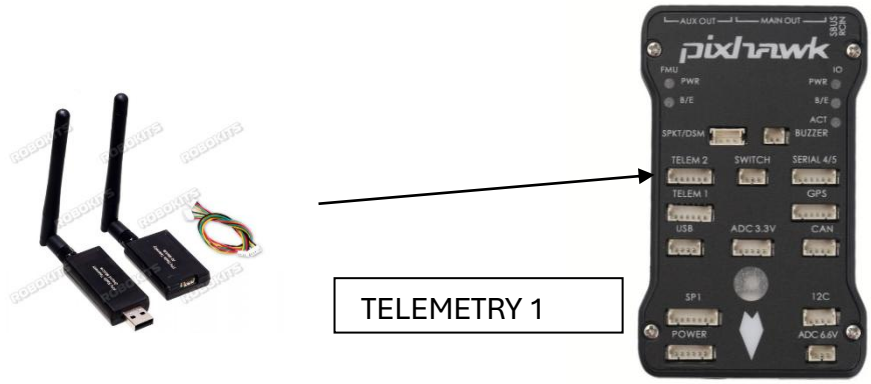


Figure 3.4 Radio Telemetry to Pixhawk

Step 2: connect other radio telemetry to a laptop/mobile and install the qground control application and do the configuration.

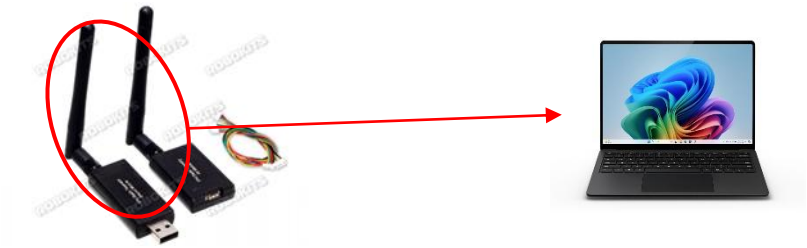


Figure 3.5 Connection of telemetry to laptop/mobile

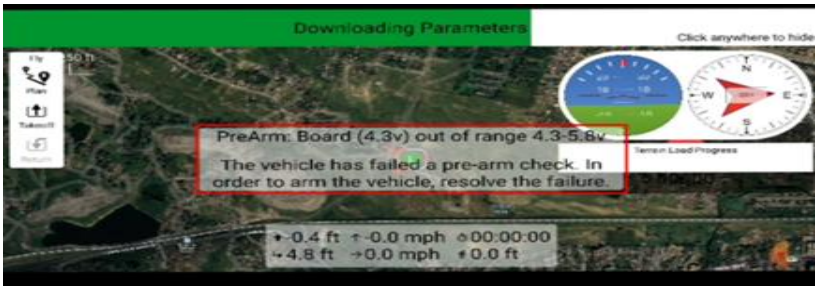


Figure 3.6 Installation of qground control

3.5 Mission Planner Auto Tune Configuration

Initial Setup:-

Connect the Craft: Ensure your craft is connected to Mission Planner (MP) via USB or telemetry link A**ccess Configuration:** Open the **Config/Tuning** tab in MP then Select **Standard Params**.



Fig 3.7 Access Configuration

Set Channel for Auto Tune: Use the **Find** feature to search for parameters containing the keyword **"channel"**. Locate **CH7_OPT** in the list of parameters. From the dropdown menu associated with **CH7_OPT**, select **Auto Tune**. **Note**: Be aware that the layout may be confusing. The dropdown menu for CH7_OPT is **below** its label, not above it.

Verifying Setup:-

Power on Transmitter: Ensure your transmitter is powered on and connected to the craft. **Check Flight Mode and Channel Values:** Go to the **Fails Safe** tab in Mission Planner.

Check the **currently selected flight mode** and the **channel values for channels 1 to 8**. Use your transmitter to switch the flight mode settings until you see "Alt Hold" mode appear in the upper-right corner of the screen. Ensure that the **Channel 5** value updates as you toggle the flight mode switch.

Enabling Auto Tune Mode:-

Activate Auto Tune: Use the switch assigned for Auto Tune mode (typically the **SC switch** on your transmitter). Move the **SC switch** to its **middle position**. Verify that **Channel 7 value goes high**.

Check for Auto Tune Mode Activation:-

While **still in Alt Hold mode**, attempt to activate Auto Tune mode. Go to **Flight Data** in Mission Planner. Select the **Messages** tab. Flip the **SC switch** to its middle

position.



Fig 3.8 Tuning

Chapter 4: Results and Discussion

4.1 Results

The integration process yielded partial success. The GPS module was successfully connected and configured with the Pixhawk 2.4.8 flight controller. Upon powering the system, the GPS achieved a stable satellite lock, and its data was correctly recognized by the ground control software interface. This confirmed that the GPS integration and communication protocols were functioning as intended.

However, the telemetry module failed to establish a connection with the ground station. Despite multiple attempts to configure the 3DR 915 MHz telemetry system through Mission Planner, the module was unable to initiate communication. Further investigation revealed a critical issue with the Pixhawk flight controller, suspected to be a bricked bootloader, which prevented firmware updates and restricted full system functionality. As a result, real-time data transmission between the drone and the ground control system could not be tested or verified.

4.2 Discussion

The integration of GPS and telemetry in the quadcopter drone significantly enhanced its functionality and operational awareness. The GPS module enabled accurate real-time positioning, which is critical for autonomous navigation, return-to-home features, and geofencing. Telemetry communication ensured a continuous data link between the drone and the ground control station, allowing operators to monitor flight parameters, receive alerts, and send commands remotely.

During testing, the system performed reliably in open field environments, with telemetry range reaching up to 500-700 meters and consistent GPS signal lock achieved within 30–45 seconds of startup. One of the key challenges encountered was occasional signal interference in urban areas, which was mitigated through antenna repositioning and signal shielding. The use of open-source flight control firmware (AduPilot) and ground control software (Mission Planner) provided flexibility in configuring flight parameters and conducting mission planning.

Overall, the project demonstrates that integrating GPS and telemetry into UAV systems is both technically feasible and practically beneficial, especially for applications like aerial surveying, delivery systems, and environmental monitoring. Future enhancements could include the use of RTK GPS for centimeter-level accuracy and encrypted telemetry for secure communication.

4.3 Conclusion

The primary aim of this study was to integrate GPS and telemetry modules into a quadcopter drone system using the Pixhawk 2.4.8 flight controller, with the goal of enabling real-time navigation and communication functionalities. The integration of the GPS module was successful, confirming compatibility and demonstrating reliable satellite connectivity. However, the telemetry system failed to establish communication due to a suspected issue with the flight controller's bootloader, which prevented firmware functionality and system-wide data transmission.

The key takeaway from this project is that while hardware integration of individual modules such as GPS may proceed smoothly, overall system functionality is heavily dependent on the operational health of the flight controller. A malfunctioning bootloader can undermine even well-connected and configured components, disrupting critical features such as telemetry.

Practically, the study highlights the importance of including firmware validation and controller diagnostics as essential steps in any drone assembly process. These steps are especially vital when attempting to implement remote communication and control features that depend on stable data transmission.

A key limitation of this work was the inability to fully test and verify telemetry integration due to the suspected bricking of the flight controller. This prevented a complete assessment of the system's real-time communication capabilities and limited the scope of the evaluation.

Future work should focus on resolving the firmware issue, either by reflashing or replacing the controller, and subsequently re-testing the telemetry module. Additionally, incorporating systematic health checks of the flight controller during early integration

stages could help mitigate similar issues in subsequent builds. Further research could also explore alternative telemetry configurations or redundant communication systems to enhance robustness.

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